

Do Variational Quantum Circuits Help Physics-Informed Neural Networks?

A Systematic Ablation Study on a Diffusion PDE Benchmark

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Abstract

Physics-informed neural networks (PINNs) that replace part of the network with a variational quantum circuit (QCPINN) have recently been reported to match or exceed classical PINNs while using far fewer trainable parameters. We evaluate this claim on a seawater temperature diffusion PDE with three boundary-condition types (Dirichlet, Neumann, Robin) and both forward and inverse problem variants, using the quantum circuit ansatz proposed by Farea et al. [2025]. Across all six scenarios, the quantum branch is both less accurate ($1.5\text{--}14.4\times$ higher mean absolute error) and substantially slower in wall-clock time ($53\text{--}89\times$) than the classical baseline. Through a two-stage ablation – first over training hyperparameters (initialization scale, learning-rate schedule), then over circuit structure (qubit count, circuit depth) with direct measurement of the circuit-only gradient norm – we rule out hyperparameter choice as the cause and find a structure-dependent effect that fits neither a clean expressivity-limited nor a clean barren-plateau account. We discuss why wall-clock accounting, largely absent from prior QCPINN evaluations, changes the practical picture, and what our gradient measurements add to existing theoretical barren-plateau claims.

1 Introduction

One-dimensional temperature diffusion is not only an ocean-scale phenomenon: the same governing equation describes heat transport in engineered water bodies such as swimming pools, where heating and cooling account for a significant share of operating energy costs, and a fast, differentiable surrogate for the temperature field is potentially useful for energy-aware control. Physics-informed neural networks (PINNs) [Raissi et al., 2019] solve forward and inverse partial differential equation (PDE) problems of exactly this kind by embedding the governing equation, boundary conditions, and (for inverse problems) observational data directly into the training loss of a neural network surrogate. We study this class of problem through a specific, well-characterized instance: the seawater temperature diffusion equation of Han, under Dirichlet, Neumann, and Robin boundary conditions, in both forward (diffusivity known) and inverse (diffusivity estimated from sparse data) form. This equation is a useful benchmark precisely because closed-form analytical solutions exist for all three boundary-condition types, which lets us measure error against ground truth exactly rather than against a numerical reference solution – a property we rely on throughout this paper and return to in Section 3.

Quantum machine learning (QML) has, in recent years, pushed to connect itself to problems beyond textbook physics – optimization, materials discovery, and increasingly scientific and engineering simulation of exactly the kind PINNs target. Hybrid quantum-classical PINNs (QCPINN/QPINN), which replace part of the classical network with a variational quantum circuit, are the concrete form

this push has taken for PDE-constrained learning: recent work reports that such hybrids match or exceed classical PINN accuracy while using a fraction of the trainable parameters [Farea et al., 2025]. If this line of work is to matter for applied surrogate-modeling settings like the energy-management example above, rather than only for problems chosen because they are convenient for a quantum circuit to represent, it needs to be evaluated with the same accounting for training cost that a practical deployment would require – not accuracy or parameter count alone.

Despite growing enthusiasm for hybrid quantum-classical PINNs, we find that efficiency in this literature is reported almost exclusively via parameter count or number of training epochs, rarely via wall-clock time – a distinction that matters because variational quantum circuits are, in the near term, executed via classical simulation whose per-step cost scales with circuit width and depth (see Section 2). This raises two questions that motivate the present study:

RQ1 On a benchmark PDE with exact analytical solutions, does a QCPINN built on a previously proposed circuit ansatz [Farea et al., 2025] match classical PINN accuracy once wall-clock training time is accounted for, not just parameter count?

RQ2 If it does not, is the gap attributable to training hyperparameters (initialization, learning-rate schedule), or to the quantum circuit’s structure (expressivity or trainability)?

We make the following contributions:

- A controlled classical-vs-quantum comparison across six matched scenarios (three boundary-condition types $\times$ forward/inverse), with explicit wall-clock accounting alongside accuracy metrics (Section 4).
- A two-phase ablation that first rules out training hyperparameters as the cause of the quantum branch’s underperformance, then isolates circuit structure (qubit count, depth) while directly measuring the circuit-only gradient norm – a diagnostic largely absent from prior QCPINN evaluations, which infer trainability issues indirectly from accuracy curves or rely on purely theoretical gradient-decay bounds (Section 4).
- A discussion of why our findings diverge from prior QCPINN results obtained with a closely related circuit design, and what that implies for how QCPINN efficiency should be reported going forward (Section 5).

The remainder of the paper is organized as follows: Section 2 surveys prior QCPINN work in detail; Section 3 describes the PDE, the classical and quantum architectures, and the training and evaluation protocol; Section 4 presents the main classical-vs-quantum comparison and the two-phase ablation; Section 5 interprets these results against prior work and states limitations; Section 6 concludes.

2 Related Work

Physics-informed neural networks (PINNs) [Raissi et al., 2019] have become a standard tool for solving forward and inverse PDE problems by embedding the governing equations directly into the training loss. Motivated by broader interest in quantum machine learning for scientific computing, a growing line of work replaces part of the classical network with a variational quantum circuit (VQC), typically in a hybrid architecture where classical layers handle pre- and post-processing while the VQC provides the trainable core – an approach variously termed QPINN or QCPINN.

Farea et al. [2025] introduce QCPINN and evaluate it, alongside a continuous-variable (qumode-based) variant that trains very poorly across the board, on five forward-only PDE benchmarks with Dirichlet-type boundary conditions: Helmholtz, a 2D lid-driven cavity flow, a 1D wave equation, Klein-Gordon, and convection-diffusion. Their headline claim – 10–30% of the trainable parameters of a classical PINN with comparable or better accuracy – holds for three of the five: a 4–64% reduction in relative L_2 error on Helmholtz, Klein-Gordon, and convection-diffusion. On the remaining two, the quantum branch is *less* accurate than the classical baseline despite using far fewer parameters (cavity: 11–23% higher relative error across the three output fields; wave: 12% higher), which the authors state plainly rather than omit [Farea et al., 2025, Sec. 8, “Conclusion”]. They also report a timing column absent from the abstract: per-iteration wall-clock time is 15–40 $\times$ that of the classical baseline in every one of the five benchmarks, which they attribute to the cost of simulating the circuit classically rather than to the algorithm itself. Separately, their qubit-based circuit sweep tests four named topologies (Alternate, Cascade, Cross-mesh, Layered) crossed with two data-encoding schemes (amplitude, angle); the accuracy and timing numbers above are for each benchmark’s best-performing topology, most often Cascade or Cross-mesh – both of which use continuously-tunable controlled-rotation entanglers (CRX/CRZ) rather than fixed CNOT gates, which the authors argue is why those two topologies converge more reliably than Alternate or Layered.

Leong et al. [2025] report a closely related pattern on a different problem family: the compressible Euler equations, tested across a smooth harmonic traveling wave, a non-harmonic solution with a moving contact discontinuity, and a 2D transonic aerofoil flow with a shock. A quantum branch whose functional form is an explicit truncated Fourier series achieves the single best loss of any model on the harmonic case (an order-of-magnitude improvement over the best classical network, using far fewer parameters), but “significantly underperformed” on the non-harmonic case and underperformed further still on the shock-dominated transonic case, where the quantum branch is additionally under-parameterized. The paper’s own stated thesis is that harmonicity of the target solution, not problem difficulty in general, is what determines whether the quantum branch helps: “quantum PINNs excel in harmonic regimes... due to the Fourier structure of PQCs... [but] struggle to generalize to discontinuities or shocks.” On timing, wall-clock cost is reported only for the transonic case, where per-epoch cost for the quantum configurations ranges 17–82 $\times$ that of the classical configurations tested (the paper’s own summary rounds this to “up to 100 $\times$ ”); a separate, larger “two orders of magnitude” figure appearing earlier in the paper is attributed to other authors’ measurements, not to this paper’s own experiments. Notably, this paper does not use the term “barren plateau” anywhere; its account of trainability loss in larger quantum models is framed instead as “excessive expressibility,” a distinct (if related) diagnosis.

Read together, these two papers already show that the value of a quantum branch in a hybrid PINN is problem-dependent rather than uniform – within a single paper’s own benchmark suite, in both cases – and that reporting practices vary: parameter-count efficiency is reported everywhere, but wall-clock or per-iteration time is reported only for a subset of benchmarks in each paper (all five in Farea et al., 2025, only the hardest of three in Leong et al., 2025), which limits how directly either paper’s timing claims generalize.

A third study applies the same qubit-based circuit family to a different application domain: Rao et al. [2025] solve four reservoir- engineering PDEs – a linear pressure-diffusion equation, the nonlinear Buckley-Leverett saturation-front equation, a linear convection-diffusion- adsorption equation, and a fully coupled nonlinear two-phase pressure- saturation system – with the same three named circuit topologies (Cascade, Cross-mesh, Alternate) as Farea et al. [2025]. This is not an independent replication: the authors state directly that their circuit implementation “is developed based on the open-source code by Farea et al. [2025],” reusing the same architecture

skeleton for a new problem class rather than testing it from a separate implementation. Two findings distinguish this study’s claim from the parameter-efficiency narrative associated with that lineage. First, parameter counts are deliberately matched rather than reduced (e.g. 273 classical vs. 279–283 quantum parameters in their simplest case) “to ensure fairness in the comparison” – the reported accuracy gain is thus an equal-budget comparison, not a parameter-efficiency one, despite reusing the QCPINN label. Second, the advantage is not uniform across topologies or benchmarks: no single topology wins on all four PDEs, the authors themselves note the Cross-mesh topology “does not consistently yield superior results across all scenarios,” and one benchmark’s own figure shows a quantum configuration’s error matching or exceeding the classical baseline’s at a specific time step, even as the paper’s overall framing is positive. All four benchmarks are forward-only with Dirichlet and Neumann-type (no-flow) boundary conditions – no inverse problems, and no Robin conditions. Notably, wall-clock or per-iteration training time is not reported anywhere in this paper, not even as a partial column of the kind Farea et al. [2025] and Leong et al. [2025] include for some subset of their benchmarks – the only acknowledgment of quantum simulation cost is a single qualitative sentence noting that “quantum simulation has inherent overhead on classical hardware,” with no accompanying measurement.

A fourth paper is worth a brief, separately-caveated mention rather than treatment as a fourth equivalent data point. Hewage et al. [2026] propose a hybrid quantum-classical architecture, independently designed rather than built on Farea et al. [2025]’s implementation, and apply it to Sri Lankan flood data using the Saint-Venant shallow-water equations. Despite the framing, the underlying task is a four-class flood-severity *classifier* with the PDE residual used only as a loss regularizer, not a solution field scored against ground truth – so its headline “71.8% vs. 69.4%” comparison is classification accuracy, not a relative-error metric, and is not commensurable with the accuracy numbers reported by any of the other three papers or by our own study. The paper’s own results table also contextualizes that comparison considerably: both physics-constrained models trail the same authors’ non-physics classical baselines (random forest 90.3%, Bayesian neural network 91.96%) by roughly 20 points, sit below what a majority-class classifier would score given a 91% class imbalance, and a transfer-learned variant of their own quantum model outperforms the flagship configuration being compared (73.6% vs. 71.8%). Like Rao et al. [2025], this paper reports no wall-clock or per-iteration timing anywhere, so its claimed “2.2–3.6× fewer epochs to converge” cannot be checked against, and may not survive, real training-time accounting – a fourth instance of the same reporting gap. Its barren-plateau-mitigation argument rests on a single gradient- variance measurement at one fixed qubit count rather than the scaling curve the underlying theory (which it does not connect to Holmes et al., 2022) would need to substantiate.

On the purely theoretical side, Lantigua et al. [2025] propose a coupling scheme between classical and quantum branches and prove that, under it, the quantum-parameter gradient norm is bounded above by $\mathcal{O}(1/\sqrt{\mathcal{N}N})$ in circuit depth $\mathcal{N}$ and qubit count N – a polynomial rather than exponential decay – with no PDE experiments of any kind (the paper’s own conclusion lists “experimental validation on both ordinary and partial differential equations” as future work). Two caveats limit how much weight this result can carry as context for our own findings. First, the architecture as formulated handles ordinary, not partial, differential equations: its network takes a single scalar input t , not the multiple independent variables (e.g. our (z, t)) a PDE requires. Second, the specific polynomial rate is obtained by first choosing the target approximation tolerance to scale as $1/\sqrt{\mathcal{N}N}$ and then showing the gradient bound matches that same rate, which reads as a property of that choice rather than an emergent, ansatz-intrinsic result – we cite the qualitative conclusion (polynomial rather than exponential decay is achievable under some coupling schemes) rather than the specific rate.

Our work differs from these four closely related studies in three ways. First, three of the

four restrict the quantum branch to forward PDE-solving problems with Dirichlet and, at most, Neumann-type boundary conditions (Farea et al., 2025 and Rao et al., 2025 exclusively forward; Leong et al., 2025 likewise), while the fourth [Hewage et al., 2026] is not a PDE-solving comparison at all in the sense used here; we test both forward and inverse formulations across three boundary-condition types (Dirichlet, Neumann, Robin) on a single diffusion PDE with closed-form solutions in all six cases, which lets us measure error against ground truth exactly rather than against a numerical or experimental reference, or a downstream classification label. Second, wall-clock accounting is inconsistent or absent across all four: Farea et al. [2025] report per-iteration time as one column among many, Leong et al. [2025] report it for only one of three benchmarks, and neither Rao et al. [2025] nor Hewage et al. [2026] report any timing measurement at all despite the former explicitly acknowledging that simulation overhead exists and the latter reporting an epoch-count “speedup” with no way to check whether it survives real-time accounting; we report wall-clock training time as a primary outcome for every scenario alongside accuracy, and find the quantum branch uniformly worse on both axes in all six – a stronger and more consistent negative result than any prior paper’s own (partially mixed, in the case of the first two) findings. Third, none of the four isolates circuit structure as an experimental variable with direct gradient measurement: Farea et al. [2025] discusses trainability differences between entangler choices (CRX/CRZ vs. fixed CNOT) qualitatively, and formally invokes barren plateaus only for its continuous-variable circuit, which is not the qubit-based design we build on; Leong et al. [2025] attributes reduced trainability in larger circuits to “excessive expressibility” based on final-loss curves; Rao et al. [2025] offers only qualitative loss-curve comparisons and explicitly disclaims any causal circuit-selection rule; Hewage et al. [2026] offers a single gradient- variance measurement at one fixed qubit count rather than the qubit-count-scaling curve its own theoretical claim would need. We instrument the circuit-only gradient norm directly across a controlled qubit-count/depth sweep, providing an empirical data point none of the four papers’ diagnostic approaches supplies. We also note, for framing rather than as a fourth point of contrast, that Rao et al.’s study is a direct architectural descendant of Farea et al.’s open-source implementation rather than an independent replication; the convergent finding across it and Leong et al. [2025] – that a quantum branch’s advantage, where present, is topology- or regime-dependent rather than uniform – is nonetheless drawn from two separate author teams and problem domains.

3 Method

3.1 Problem setup

We consider the one-dimensional temperature diffusion equation

$$\frac{\partial T}{\partial t} = k_z \frac{\partial^2 T}{\partial z^2}, \quad z \in [0, 2], \quad t \in [0, 5], \quad (1)$$

with true diffusivity $k_z = 0.05$, following han. We evaluate three boundary conditions, each with a closed-form analytical solution used as ground truth:

- **Dirichlet:** $T = e^{-k_z(\pi/2)^2 t} \sin(\pi z/2) + e^{-k_z \pi^2 t} \sin(\pi z)$.
- **Neumann:** the corresponding cosine-series solution.
- **Robin:** $\partial T/\partial z + T = 0$ at $z = 0, 2$; the general-term solution follows han, Eq. 18.

For each boundary condition we solve both a *forward* problem (k_z known, solve for T) and an *inverse* problem (k_z unknown, estimated jointly with T from sparse observations), giving six scenarios in total.

3.2 Classical PINN

The classical baseline is a fully connected network mapping $(z, t) \mapsto T$: five hidden layers of 32 units with tanh activations. The loss combines the PDE residual, boundary-condition residual, and (for inverse problems) a data-fit term against sparse noisy observations; k_z is a trainable scalar in the inverse case (parameterized as $\exp(\log k_z)$ for positivity). Training samples 2540 PDE-residual (collocation) points, 80 boundary points (40 per side), and 160 initial-condition points, each redrawn uniformly at random per run; inverse scenarios additionally use 12 noiseless observation points on a fixed 4×3 (z, t) grid. The loss term weights are $w_{bc} = w_{ic} = w_{data} = 1$ throughout, and $w_{pde} = 1$ for Dirichlet and Neumann scenarios; Robin scenarios use $w_{pde} = 10$, since the gradient-dependent Robin boundary condition is harder to satisfy at $w_{pde} = 1$ and needs the PDE residual up-weighted to converge (an asymmetry inherited from the original classical replication rather than introduced for this study). Training is two-phase: Adam ($\text{lr} = 10^{-3}$) for a fixed number of steps (5000 for Dirichlet/Neumann, 20000 for Robin, matching the higher w_{pde}), followed by L-BFGS (strong-Wolfe line search, history size 100, $\text{tolerance_grad} = 10^{-9}$, $\text{tolerance_change} = 10^{-11}$, up to 10000 or 20000 iterations matching the Adam budget) until L-BFGS’s own convergence tolerances are met or the iteration budget is exhausted, whichever comes first – there is no early stopping based on held-out validation error.

3.3 QCPINN

The quantum branch is inspired by, but not identical to, the qubit-based (discrete-variable) circuit family in Farea et al. [2025]. Inputs (z, t) are linearly rescaled to $[-1, 1]$ and encoded via **AngleEmbedding** (RX rotations) onto n qubits, matching their angle-embedding scheme. Each of L variational layers applies RZ-RX rotations on every qubit, a *ring* of CNOT entangling gates (qubit i to qubit $(i + 1) \bmod n$, wrapping from the last qubit back to the first), then a mirrored RX-RZ rotation layer; the circuit output is the vector of Pauli-Z expectation values $\langle Z_i \rangle \in [-1, 1]$, matching their readout. Farea et al. [2025] test four named topologies, none of which is exactly this combination: their *Cascade* topology uses ring connectivity but continuously-tunable CRX gates rather than fixed CNOT, while their *Alternate* and *Layered* topologies use fixed CNOT but nearest-neighbor (not ring) connectivity. We therefore refer to our circuit as *ring-CNOT*: a fifth ring-plus-fixed-entangler combination that is closest in spirit to their Cascade topology but was not itself among the four they swept, rather than an instance of any one of their named topologies. This output is post-processed by a small classical head to produce T (and, for inverse problems, k_z). Circuit parameters are simulated exactly (no shot noise, no hardware noise) via PennyLane’s `default.qubit` device with the `backprop` differentiation method – Farea et al. [2025] state they use the same shots=None/backprop configuration for their qubit-based circuits but do not print the specific simulator device string, so this match is inferred, not confirmed verbatim. The baseline configuration uses $n = 4$ qubits and $L = 2$ layers; note this differs from the $n = 5$, $L = 1$ configuration Farea et al. [2025] found best after their own sweep – Section 4 sweeps qubit count and depth independently around our own baseline, not theirs. The quantum branch is trained differently from the classical one, not only architecturally but in optimizer choice: it never switches to L-BFGS – each L-BFGS step would require multiple closure evaluations through the quantum circuit, which is too expensive per step – and instead runs Adam throughout, for a step budget equal to the classical branch’s combined Adam+L-BFGS budget (i.e. 15000 steps for Dirichlet/Neumann, 40000 for Robin), with cosine learning-rate decay from 10^{-3} to a floor $\eta_{\min} = 10^{-5}$. This means the two branches differ in optimizer (first-order throughout vs. first-order then second-order) as well as in network type, a confound Section 4’s phase-1 ablation partially addresses by varying $\eta_{\min}$ but does

not eliminate – both branches were given a matched total step/iteration budget, not a matched optimizer.

3.4 Training and evaluation

All experiments use PyTorch with fixed seeds. Seeding differs by experiment: the main classical-vs-quantum comparison (Section 4, Table 1) uses a single seed (0) per scenario, so the reported numbers there are point estimates with no variance across runs; the phase-1 and phase-2 ablations each use 3 seeds (0, 1, 2) per (scenario, configuration) pair, and it is only for those that we report a mean and standard deviation. We return to what this asymmetry means for how the main comparison should be read in Section 5’s limitations. We report:

- **MAE** and **relative L_2 error** against the analytical solution on a held-out grid;
- **PDE residual** at collocation points, as an optimization-quality check independent of the analytical solution;
- k_z **estimation error** for inverse scenarios;
- **Wall-clock training time**, measured on an Apple M2 (8 cores, 16GB RAM), CPU only for both branches – neither PyTorch nor PennyLane was given GPU access, so the reported ratios are a CPU-only comparison. This likely does not favor the classical branch disproportionately: a GPU-accelerated classical network would train faster still, which would widen rather than narrow the gap, whereas the quantum branch’s cost is dominated by classical simulation of the circuit itself and would not benefit comparably from the same hardware upgrade;
- **Circuit-only gradient norm** (phase-2 ablation only): the norm of the loss gradient with respect to the VQC parameters alone, logged at the start and end of training, isolating circuit trainability from the classical pre/post-processing layers.

4 Experiments

4.1 Classical vs. quantum: main comparison

Table 1 compares the classical and quantum branches across all six scenarios. The quantum branch is less accurate in every scenario (1.5–14.4× higher MAE) and substantially slower in every scenario (53–89× wall-clock time), with no scenario favoring the quantum branch on either axis.

Figure 1 shows this gap spatially for the Dirichlet-forward scenario: the classical solution tracks the analytical field closely across the full (z, t) domain, while the quantum solution’s larger error is visible directly rather than only as an aggregate number.

4.2 Ablation 1: training hyperparameters

The main comparison leaves open whether the quantum branch’s underperformance reflects a suboptimal training recipe rather than a structural limitation. We test two candidates on two representative scenarios (Dirichlet-forward, Robin-inverse): a smaller weight initialization scale (`small_init`) intended to keep VQC rotation angles away from regions with vanishing gradient, and a higher learning-rate floor (`slow_decay`) intended to prevent premature convergence to a poor optimum. Table 2 shows neither closes the gap: `slow_decay` is the worst of the three quantum configurations in both scenarios, and `small_init` yields only a 3–16% MAE improvement over the untouched baseline – far short of the 3–14× gap to the classical branch.

Table 1: Classical vs. quantum PINN across six boundary-condition / problem-type scenarios. MAE and wall-clock time (s) are means over seeds; ratio columns are quantum relative to classical (higher = quantum worse).

BC	Problem	MAE (cls.)	MAE (qtm.)	MAE ratio	Time (cls.)	Time (qtm.)	Time ratio
Dirichlet	Forward	0.000231	0.00034	1.5×	31.9	2.44e+03	76.5×
Dirichlet	Inverse	0.000287	0.0005	1.7×	39.1	2.69e+03	69.0×
Neumann	Forward	0.000166	0.00239	14.4×	38.4	2.56e+03	66.6×
Neumann	Inverse	0.000249	0.00118	4.7×	35.6	2.82e+03	79.3×
Robin	Forward	8.96e-05	0.000384	4.3×	128	6.79e+03	52.9×
Robin	Inverse	8.66e-05	0.00029	3.3×	133	1.19e+04	89.3×

Table 2: Phase-1 ablation: quantum-branch initialization scale and LR-schedule floor, vs. the classical reference and the untouched quantum baseline. Neither intervention closes the accuracy or runtime gap.

Scenario	Config	MAE (mean $\pm$ std)	Time (s)
Dirichlet Forward	classical	0.000171 $\pm$ 3.7e-05	65.68
Dirichlet Forward	baseline	0.000522 $\pm$ 0.00013	5257
Dirichlet Forward	small_init	0.000437 $\pm$ 6e-05	5097
Dirichlet Forward	slow_decay	0.00102 $\pm$ 0.0008	5038
Robin Inverse	classical	7.91e-05 $\pm$ 2e-05	400.3
Robin Inverse	baseline	0.000626 $\pm$ 0.00033	1.492e+04
Robin Inverse	small_init	0.000609 $\pm$ 0.00023	1.513e+04
Robin Inverse	slow_decay	0.0011 $\pm$ 0.00032	1.476e+04

4.3 Ablation 2: circuit structure

Having ruled out these two hyperparameters, we next ask whether the circuit itself is the bottleneck – either under-expressive (too few qubits/layers) or subject to a barren-plateau-like effect (gradients vanishing as the circuit grows). We sweep qubit count and depth independently around the baseline ($n=4, L=2$): **narrow** ($n=2$), **wide** ($n=6$), **deep** ($L=4$), holding initialization and learning-rate schedule at the phase-1 baseline throughout. Alongside the usual accuracy metrics we log the circuit-only gradient norm at the end of training. Table 3 summarizes the result.

Figure 2 shows the same pattern the table does, but makes the non-monotonicity easier to see at a glance.

The pattern is mixed rather than clean. **wide** achieves the best accuracy among the quantum configurations in both scenarios, which would suggest expressivity is a limiting factor worth scaling past – but it also has the *smallest* final gradient norm of the four configurations, which is not the direction a pure expressivity story predicts and is only partially consistent with a barren-plateau account (which predicts smaller gradients accompanying *worse*, not better, accuracy). **narrow** is the worst-performing configuration on accuracy, consistent with an under-expressivity explanation at the low end, while **deep** is worse than baseline on accuracy despite a comparable gradient norm. Taken together, no single one of these three configurations’ results is consistent with the other two under either account on its own: an expressivity-limited story explains **narrow** but not **wide**’s small gradient norm, and a barren-plateau story explains **wide**’s small gradient norm but not its best-in-

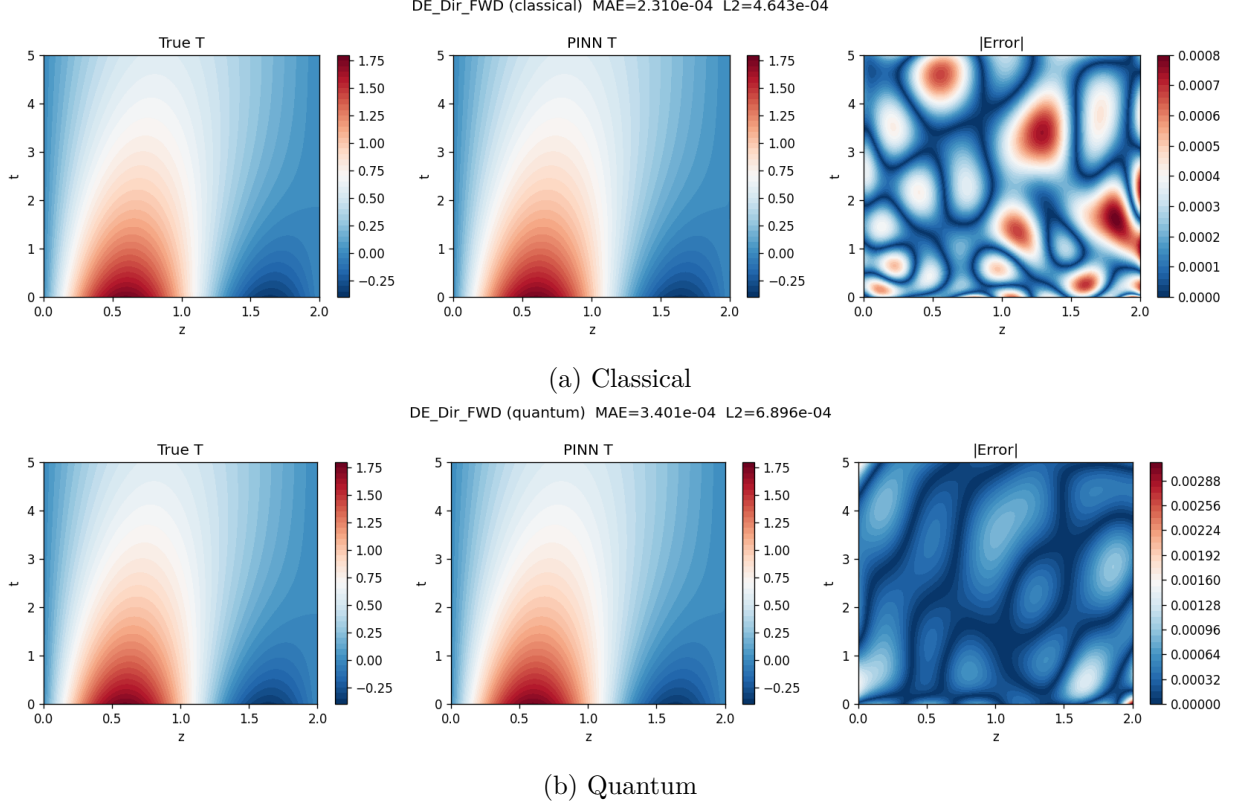


Figure 1: Predicted solution, analytical solution, and pointwise error for the Dirichlet-forward scenario, classical (top) vs. quantum (bottom).

class accuracy. We therefore do not adopt either explanation as a conclusion; Section 5 discusses what would be needed to adjudicate between them, and what other factor – data encoding, ansatz choice – the sweep as designed cannot rule in or out.

5 Discussion

5.1 Prior positive results are narrower than they first appear

None of Farea et al. [2025], Leong et al. [2025], or Rao et al. [2025] is a uniform “quantum wins” result once read past the abstract: all three contain within-paper negative or mixed cases – cavity flow and the wave equation for the first, non-harmonic and transonic regimes for the second, no circuit topology winning across all four benchmarks (plus an explicit disclaimer against causal claims) for the third. We read our six-scenario result not as contradicting these papers but as a more consistent, more uniform version of a pattern they already show in part: where they find a mix depending on PDE, regime, or topology, we find the quantum branch losing on every single one of six matched scenarios on a problem family – a diffusion equation under three boundary-condition types, in both forward and inverse form – that none of the three tests at all: all three restrict to forward problems with Dirichlet (and, in one case, additionally Neumann) conditions, and none includes an inverse/parameter-estimation variant. It is also worth being precise about how much independent weight Rao et al. [2025] carries: it is not an arms-length third data point, since it explicitly reuses Farea et al.’s open-source implementation for a new PDE domain. Its convergence

Table 3: Phase-2 ablation: circuit width/depth sweep (narrow/baseline/wide/deep) with final circuit-only gradient norm. Accuracy and gradient norm do not move monotonically together, so neither a pure expressivity nor a pure barren-plateau account fits cleanly – see Discussion.

Scenario	Config	MAE (mean $\pm$ std)	Final $\ \nabla_{\text{circuit}}\ $	Time (s)
Dirichlet Forward	classical	$0.000184 \pm 3.5\text{e-}05$	0	31.96
Dirichlet Forward	baseline	0.00052 ± 0.00013	$1.56\text{e-}05$	2655
Dirichlet Forward	narrow	0.00152 ± 0.00052	$1.93\text{e-}05$	968.1
Dirichlet Forward	wide	$0.000446 \pm 8.9\text{e-}05$	$9.45\text{e-}06$	7398
Dirichlet Forward	deep	0.000663 ± 0.00023	$2.21\text{e-}05$	4431
Robin Inverse	classical	$6.44\text{e-}05 \pm 1.6\text{e-}05$	0	151.3
Robin Inverse	baseline	0.000627 ± 0.00033	0.00653	8331
Robin Inverse	narrow	0.00177 ± 0.00092	0.0326	3173
Robin Inverse	wide	0.000413 ± 0.00013	0.00575	$2.201\text{e}+04$
Robin Inverse	deep	0.000508 ± 0.0001	0.00595	$1.393\text{e}+04$

with Leong et al. [2025] on the qualitative claim that a quantum branch’s advantage is regime- or topology-dependent rather than uniform therefore reflects two independent author teams agreeing on that shape, not three independent confirmations of any specific number.

5.2 Wall-clock accounting: sharper than the literature, but comparable

Farea et al. [2025] report per-iteration time (15–40 $\times$ slower than classical, across all five of their benchmarks) and Leong et al. [2025] report per-epoch time for one of their three benchmarks (17–82 $\times$, which they themselves round to “up to 100 $\times$ ”). Rao et al. [2025] and Hewage et al. [2026] sit at the far end of this spectrum: the former runs on a fully specified CPU (two Intel Xeon E5-2680 v4, 28 cores, 128GB RAM) and explicitly acknowledges in prose that “quantum simulation has inherent overhead on classical hardware,” yet reports no timing measurement whatsoever; the latter reports an epoch-count “speedup” (fewer epochs to reach a validation-loss threshold) without ever mentioning wall-clock cost at all, despite using parameter-shift-rule gradients – known to require multiple extra circuit evaluations per step relative to backprop – which makes an epoch-count comparison an especially weak proxy for real training cost in that specific case. Two of four closely related papers reporting zero timing data, and a third reporting it for only one of three benchmarks, is itself worth stating as a field-wide pattern rather than a coincidence across independent author teams. Our own 53–89 $\times$ wall-clock ratio is in the same direction as the two papers that do measure it, and a similar order of magnitude – not a qualitatively new finding – but we measure it as a primary outcome across all six of our scenarios rather than as a secondary column reported for some subset of benchmarks, or omitted entirely. Our ratio runs somewhat higher than Farea et al.’s 15–40 $\times$; one plausible source is circuit size, since their best-performing configuration uses only 5 qubits and 1 layer, smaller than our 4-qubit, 2-layer baseline (and our own ablation sweeps up to 6 qubits and 4 layers, which should be commensurately slower still). However, none of the three papers states enough hardware detail to rule out a hardware difference as a contributing factor instead or in addition – Farea et al. report a 6148-CPU, 192GB cluster node; the flow paper reports only “distributed CPU, no GPU”; the reservoir paper specifies its CPU but has no timing to compare against at all – so this is a limit on how precisely wall-clock ratios can be compared *across* papers, not only a feature of our own results.

5.3 A smooth problem that still loses – and why “barren plateau” needs care

Leong et al. [2025]’s central thesis is that harmonicity of the target solution predicts whether the quantum branch helps, not problem difficulty in general. Our diffusion PDE has closed-form sine/cosine solutions and so sits on the “harmonic” side of their distinction, yet our quantum branch loses in all six scenarios, including the smoothest ones. This is a genuine point of tension worth naming directly: either the harmonic/non-harmonic axis from their compressible-flow, Fourier-series argument does not transfer to a diffusion PDE encoded via AngleEmbedding, or some other factor in our setup – data encoding, our ring-CNOT ansatz versus their tunable CRX/CRZ entanglers (Section 3), or optimizer choice – is the actual confound; our results cannot distinguish between these. We are also careful with terminology here, since neither prior paper’s account of trainability loss for the circuit family closest to ours uses “barren plateau”: Leong et al. [2025] calls it “excessive expressibility,” based on final-loss curves rather than a direct gradient measurement, and Farea et al. [2025] uses “barren plateau” explicitly but only for its continuous-variable (qumode) circuits, a different paradigm from the qubit-based ansatz both they and we build on. Our own phase-2 gradient-norm measurement (Section 4) is, as far as this literature review found, the only *direct* circuit-only gradient measurement across a qubit-count/depth sweep for a qubit-based hybrid PINN. We offer this as a contribution, but state plainly, as already noted in Section 4, that our own numbers do not cleanly resolve the underlying question either.

One theoretical result changes how that non-resolution should be framed, though: Holmes et al. [2022] show that gradient magnitude and ansatz expressibility are not independent, competing explanations but two faces of the same relationship – more expressive ansätze induce flatter cost landscapes and correspondingly smaller typical gradients. Read through that lens, **wide** having both the best accuracy *and* the smallest final gradient norm among your four quantum configurations is not the contradiction it looks like under a naive “expressivity good, vanishing gradient bad” framing – it is closer to what Holmes et al. [2022]’s bound would predict for a more expressive circuit that has not yet entered a true (exponential, system-size-dependent) barren plateau: still trainable, but with a flatter landscape than **narrow** or **baseline**. We stop short of calling this a confirmed barren plateau, however: our qubit-count range (2–6) is almost certainly too small for the asymptotic, exponential-decay regime that the barren-plateau literature [McClean et al., 2018, Holmes et al., 2022] actually characterizes. What our sweep shows is consistent with the small-system-size end of the expressibility-gradient relationship those results describe – a flatter but still-trainable landscape as the circuit widens – not evidence for or against an eventual barren plateau at larger qubit counts, which our range cannot distinguish from ordinary flattening.

5.4 Limitations

- All quantum-circuit simulation is exact and noiseless (`default.qubit`); results say nothing about performance on real NISQ hardware, where shot noise and decoherence would likely widen rather than close the gap. (Farea et al.’s feasibility analysis of their Cross-mesh circuit under a stated noise model is the closest precedent for what a hardware-realistic version of this comparison would need to account for.)
- Both branches were run on CPU. This asymmetry is worth reading carefully rather than at face value: the classical branch could in practice be GPU-accelerated, which would shrink its training time further, while the quantum branch’s cost is dominated by classically simulating the circuit and would not benefit comparably from the same hardware – so a GPU-enabled comparison would likely widen, not narrow, the wall-clock gap reported here. Under future fault-tolerant quantum hardware this asymmetry could of course reverse, but that is a state-

ment about hardware that does not yet exist, not about the noiseless-simulator comparison in this paper. We also note that neither Farea et al. [2025] nor Leong et al. [2025] give full hardware specifications sufficient for a precise cross-paper timing comparison – a field-wide reporting gap, not a limitation specific to our study.

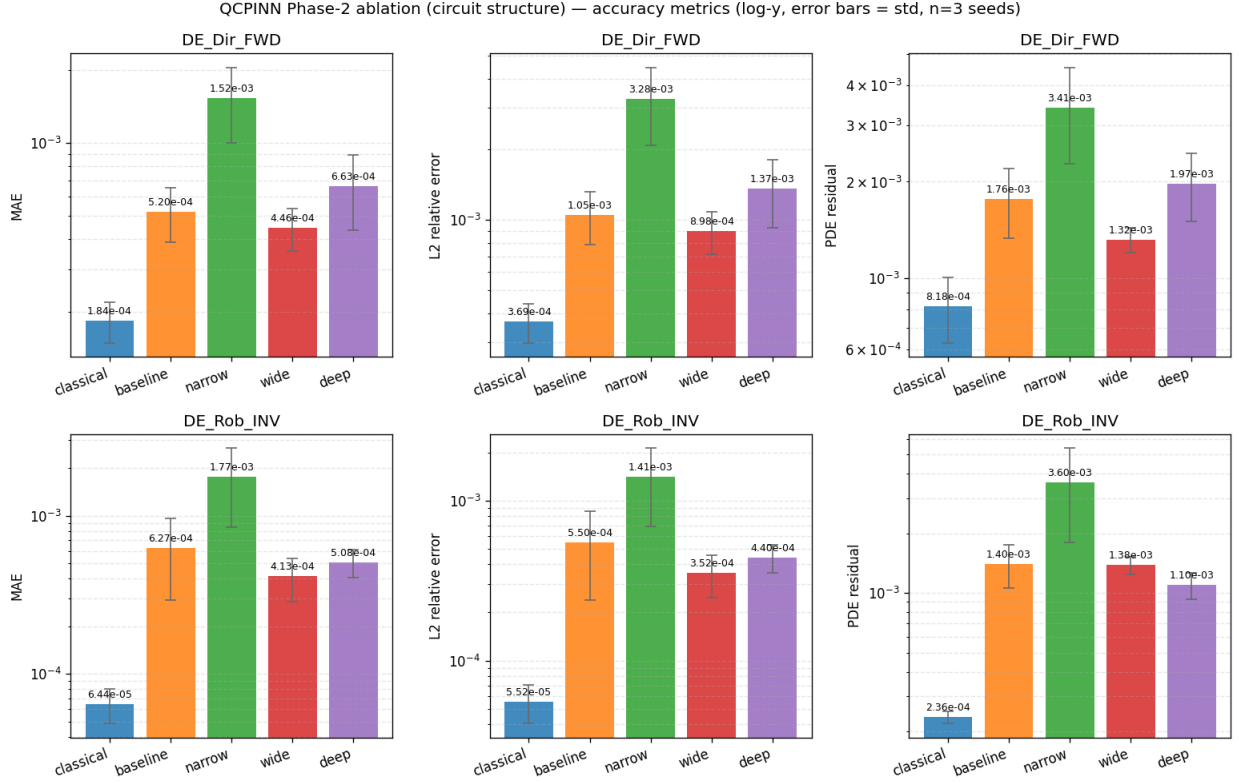
- Single PDE family (linear diffusion, 3 BC types); results may not transfer to nonlinear or higher-dimensional PDEs, or to the discontinuous/shock-type solutions where Leong et al. [2025] independently find classical dominance for different reasons.
- The two ablations use 3 seeds per (scenario, configuration), but the headline classical-vs-quantum comparison in Table 1 uses a single seed per scenario – it is the only result in the paper reported as a point estimate rather than a mean $\pm$ std, and the largest-magnitude claims (the 53–89 $\times$ wall-clock ratios) come from that single-seed table. This is a real limitation rather than a disclosure formality: the ablations show run-to-run standard deviations on the same order as some of the effects being compared, so we cannot yet rule out that the main comparison’s specific ratios would shift, within their same broad range, under a different seed. Multi-seed replication of Table 1 is the clearest concrete next step before this result is submission-ready.
- Phase-2 ablation varies qubit count and depth but not entangler type (fixed CNOT throughout) or data-encoding scheme – Farea et al. [2025] suggest entangler tunability (CRX/CRZ vs. fixed CNOT) affects convergence reliability in their own qubit-circuit sweep, which we do not test and flag as the clearest concrete next ablation.

6 Conclusion

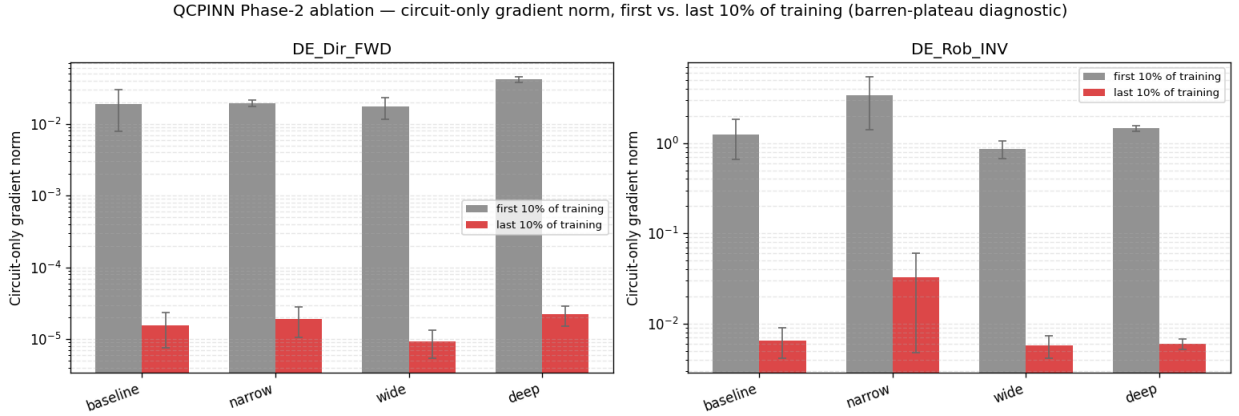
We set out to test whether a hybrid quantum-classical PINN built on a previously proposed circuit ansatz matches classical PINN performance on a benchmark diffusion PDE once wall-clock training time is accounted for alongside accuracy (RQ1), and, if not, whether the gap traces to training hyperparameters or to the quantum circuit’s structure (RQ2). It does not match classical performance: across all six boundary-condition and forward/inverse scenarios, the quantum branch is both less accurate and 53–89 $\times$ slower in wall-clock time, and this gap survives targeted hyperparameter tuning of initialization scale and learning-rate schedule. A subsequent sweep of qubit count and circuit depth, instrumented with a direct measurement of the circuit-only gradient norm, shows that circuit structure does matter, but not in a way that cleanly fits either a pure expressivity-limited or a pure barren-plateau account – the best-accuracy configuration in our sweep also has the smallest measured gradient norm, a pattern more consistent with the small-system-size end of the expressibility-gradient relationship of Holmes et al. [2022] than with either account taken alone. Beyond this specific PDE, the methodological point we think most likely to generalize is the one developed in Section 5: efficiency claims for hybrid quantum-classical PINNs in the current literature are reported almost exclusively via parameter count or training epochs, and wall-clock time is absent or incomplete in every closely related paper we examined. Our own results show that accounting for it changes the practical picture, and we suggest it become standard practice for evaluating this class of model.

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(a) MAE by configuration



(b) Final circuit-only gradient norm by configuration

Figure 2: Phase-2 circuit-structure ablation: accuracy (top) and final gradient norm (bottom) do not move together in the direction either a pure expressivity or a pure barren-plateau account would predict – **wide** has both the best accuracy and the smallest gradient norm among the quantum configurations.